

YOGESH KULKARNI

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Research Interests

My research develops **post-training methods for video-language models** that enable reasoning over long-horizon spatial, temporal, and audio-visual content. I focus on **preference alignment** and **reinforcement learning** with structured learning signals for data-efficient multimodal reasoning, including adaptive reasoning, proactive decision-making in continuous egocentric streams, and video generation as visual reasoning for spatial and physical understanding.

Education

Arizona State University, Tempe, AZ <i>Doctor of Philosophy, Computer Science</i>	Aug 2024 – May 2028
University of Southern California, Los Angeles, CA <i>Master of Science, Computer Science</i>	May 2024
Pune Institute of Computer Technology, Pune, India <i>Bachelor of Engineering, Computer Engineering</i>	July 2022

Selected Publications

- AVATAR: Reinforcement Learning to See, Hear, and Reason Over Video**
Yogesh Kulkarni, Pooyan Fazli *CVPR 2026*
- EgoVITA: Learning to Plan and Verify for Egocentric Video Reasoning**
Yogesh Kulkarni, Pooyan Fazli *ECCV 2026*
- ReGATE: Learning Faster and Better with Fewer Tokens in MLLMs**
Chaoyu Li, Yogesh Kulkarni, Pooyan Fazli *ACL Main 2026*
- Video Generation Models: A Survey of Post-Training and Alignment**
Chaoyu Li, Xiaoyi Gu, Yogesh Kulkarni, Eun Woo Im, Mohammadmahdi Honarmand, et al. *TMLR 2026*
- VideoPASTA: 7K Preference Pairs That Matter for Video-LLM Alignment**
Yogesh Kulkarni, Pooyan Fazli *EMNLP Main 2025*
- VideoSAVi: Self-Aligned Video Language Models without Human Supervision**
Yogesh Kulkarni, Pooyan Fazli *COLM 2025*

Experience

Research Scientist Intern Adobe Research, San Jose, CA Developed proactive, reasoning omni models that decide when and what to say during lecture videos, trained with GRPO and on-policy self-distillation (OPSD) on a reason-then-speak objective lifting proactive content quality to 3.1 out of 5 and recall to 0.84 on held-out lectures.	May 2026 – Present
Graduate Researcher PeRL Lab, Arizona State University Video Reasoning Models (CVPR'26), Egocentric and Procedural Video Understanding (ECCV'26), Video-Language Alignment (EMNLP'25, COLM'25), and Efficient training for MLLMs (ACL'26).	Aug 2024 – Present
Research Scientist Intern Nokia Bell Labs, New Providence, NJ Implemented automatic model parallelism for GPT-3 and LLaMA models, increasing training throughput by 15% across heterogeneous clusters. Designed communication/compute-efficient inter-node pipeline parallelism for training LLMs on geo-distributed GPUs.	May 2023 – Aug 2023
Graduate Researcher USC ICT, Los Angeles, CA Built 3D style transfer pipeline (ViT, CLIP, Gaussian Splatting) for text-guided point cloud coloring. Explored diffusion models (DDPM) and GANs (Pix2Pix, CycleGAN) with PointNet/KPConv/SparseUNet for 3D colorization.	Jan 2023 – Mar 2024

Academic & Technical Details

Languages & Frameworks: Python, C/C++, SQL, PyTorch, VerL, SWIFT, $\text{\LaTeX}$

Voluntary Service: Reviewer for NeurIPS, CVPR, ICCV, ECCV, AAAI, ARR, TPAMI, TMLR